Group Employee Insurance

The **Group Employee Insurance Management System** is a full-stack web application developed to simulate a real-world corporate (B2B) insurance platform. The system enables companies to purchase and manage group insurance policies for their employees, while allowing the insurance provider to manage products, assign agents, track commissions, and process claims through structured role-based access control using JWT authentication.

The platform operates with four primary roles: **Admin, Agent, Claims Manager, and Customer (Company HR)**. A single predefined Admin account exists within the system. The Admin is responsible for creating Agent and Claims Manager accounts and managing predefined policy templates. Customers (companies) register through the company website with the default role assigned as Customer. Upon login, users are redirected to role-specific dashboards based on their authorized role.

The system includes **nine predefined insurance plans** structured across three professional tiers: **Essential, Enhanced, and Enterprise**. Each tier contains three levels: **Base, Plus, and Pro**. The coverage structure is progressive:

- **Base** – Health coverage only
- **Plus** – Health + Life coverage
- **Pro** – Health + Life + Accident coverage

Policy tiers differ in coverage limits, life multipliers, accident coverage limits, and premium rates. These policies are predefined in the system and managed by the Admin.

Customers can view all available plans and either directly purchase a suitable policy or submit a recommendation request by providing company size and business domain details. A company is assigned to an Agent at its first policy interaction (either recommendation request or direct purchase). The assigned Agent acts as the relationship manager for that company throughout the policy lifecycle. Agents generate quotations using a predefined premium formula and earn a fixed **8% commission** on successful policy purchases made by their assigned companies.

Premium calculation follows a standardized system formula:

Premium = Base Premium Rate × Number of Employees

When a company selects a policy and requests a quote, the Agent generates the quotation based on this formula. Upon payment confirmation, the policy status is updated to **Active**, policy start and expiry dates (one-year validity) are generated, and an invoice is created and stored in the company profile. A company is allowed to maintain only **one active policy at a time**.

Companies manage their employees within the system by adding details such as name, email, annual salary, role, and nominee information. Employees can be added before or after

policy purchase. As long as the company holds an active policy, all active employees are automatically covered under that policy.

The claims module supports three structured claim types based on the purchased policy level:

- **Health Claims** – Raised for hospital expenses and allowed multiple times within the employee's remaining health coverage balance. After approval, the claimed amount is deducted from the remaining coverage and updated in the employee record.

- **Life Claims** – Calculated as Salary × Life Multiplier defined in the policy. Life claims are allowed only once per employee. Upon approval, the employee is marked as deceased and deactivated from further coverage.

- **Accident Claims** – Supported only under Pro level policies. Permanent total disability results in full accident coverage payout, while permanent partial disability is calculated based on the defined disability percentage.

Claims are submitted by the Customer on behalf of employees and require supporting document uploads. Once submitted, claims cannot be modified. All claims are reviewed by the Claims Manager, who verifies coverage eligibility and documentation before approving or rejecting the claim. Payout calculations are performed automatically by the system to maintain consistency and prevent manual manipulation.

The Admin provides system-wide oversight, including user management, policy management, company monitoring, and claim auditing. The Agent handles recommendation and quotation workflows, while the Claims Manager is responsible exclusively for structured claim validation and decision-making.

Overall, this project models a structured corporate insurance workflow incorporating secure role-based authentication, tiered product architecture, controlled agent assignment, premium calculation, invoice generation, commission tracking, employee lifecycle management, coverage-based claim validation, and centralized claim processing within a scalable B2B insurance system.

🛡️ Admin — System Administrator

1. User Management

Admin is responsible for creating operational users.

Admin can:

- Register Agent accounts
- Register Claims Manager accounts
- View all Agents
- View all Claims Managers
- View all Customers (Companies)
- Delete Customer **only if no Active Policy exists**

2. Policy Management

- Add new policy
- Edit policy (coverage, multiplier, premium rate)

- View all policies

3. Company Overview

- Company name
- Assigned Agent
- Policy taken
- Policy status (Active / Expired)
- Premium amount
- Policy start and expiry date

4. Claims Overview

Admin can:

- View all claims
- Filter by company
- See claim type
- See claim status (Pending / Approved / Rejected)
- See payout amount

Agent

Company Assignment

- A company is assigned to an Agent at its first policy interaction (Recommendation request or Direct Buy).
- Each company is linked to only one Agent.
- One Agent can manage multiple companies.

Assigned Companies Management

Agent can:

- View all companies assigned to them
- View policy status (None / Active / Expired)
- View policy details (Tier + Level)
- View premium amount
- View policy start and expiry dates

Recommendation Handling

- View recommendation requests submitted by companies
- Review company details (employee size, domain)
- Select suitable policies from predefined plans
- Add remarks or suggestions
- Send recommendation to customer

Quote Generation

Agent can:

- View selected policy
- View employee count
- Generate quotation

Premium is calculated automatically using:

$$\text{Premium} = \text{BasePremiumRate} \times \text{EmployeeCount}$$

After generating the quote:

- Agent sends it to the customer
- Customer reviews and confirms payment

Commission Tracking

Agent can:

- View purchased policies from assigned companies
- View premium amount per policy
- View fixed 8% commission earned
- View commission history
- View total commission earned

Commission is generated only after policy payment confirmation.

Claim Managers

1️⃣ View and Manage Claims

The Claims Manager can:

- View all submitted claims
- Filter claims by:
    - Company
    - Claim Type (Health / Life / Accident)
    - Status (Pending / Approved / Rejected)
    - Date

All claims appear in a centralized claims list.

2️⃣ Review Claim Details

For each claim, the Claims Manager can view:

- Company Name
- Employee Name
- Employee Salary
- Policy Name (Tier + Level)
- Claim Type
- Requested Amount
- Coverage Details
- Remaining Health Coverage (for health claims)
- System-calculated payout amount
- Uploaded supporting documents

The Claims Manager must verify:

✔ Documents are valid
✔ Claim type is covered under the policy
✔ Requested amount is within coverage limits

3️⃣ Approve or Reject Claims

The Claims Manager can:

- Approve a claim
- Reject a claim
- Add remarks or rejection reason

Once a claim is submitted by the customer, it cannot be modified.

4️⃣ System-Controlled Updates After Approval

Upon approval:

- Claim status is updated to Approved
- Health coverage balance is reduced (for health claims)
- Employee is marked deceased and deactivated (for life claims)
- Payout amount is finalized automatically
- Customer is notified

Upon rejection:

- Status is updated to Rejected
- Reason is recorded
- Customer is notified

🎯 1️⃣ Registration & Account Access

Customer can:

- Register company account
- Login to the system
- Access customer dashboard

A company is assigned to an Agent at its first policy interaction (Recommendation request or Direct Buy).

👥2️⃣ Employee Management

Customer can:

- Add employees
- Edit employee details
- Remove employees (if allowed)

Each employee record includes:

- Name
- Email
- Annual Salary
- Role
- Nominee details

Employees can be added before or after policy purchase.

If policy is Active:

✔ All active employees are automatically covered under the company's policy.

📦3️⃣ Policy Viewing & Selection

Customer can:

- View all predefined insurance plans
- View Tier (Essential / Enhanced / Enterprise)
- View Level (Base / Plus / Pro)
- View coverage details

- View per-employee premium rate

Coverage structure:

- Base → Health only
- Plus → Health + Life
- Pro → Health + Life + Accident

Customer can either:

Option A — Request Recommendation (Optional)

- Submit company size and domain
- View recommended plans from assigned Agent

Option B — Direct Buy

- Select policy directly

💰4️⃣ Quote & Policy Purchase

After selecting a policy:

Customer can:

- Enter number of employees
- Request quote

Agent generates quote using:

$$\text{Premium} = \text{BasePremiumRate} \times \text{EmployeeCount}$$

Customer can:

- View total premium
- Confirm payment

Upon payment:

- Policy status becomes Active
- Start and expiry dates generated (1-year validity)
- Invoice generated
- Commission (8%) credited to Agent
- Policy automatically applied to all active employees

Only one Active policy is allowed at a time.

📑5️⃣ Claim Submission

Customer can raise claims on behalf of employees.

Steps:

1. Select employee
2. Select claim type (based on policy level)
3. Enter requested amount
4. Upload supporting documents
5. Submit claim

Claim types allowed:

Policy Level	Allowed Claims
Base	Health
Plus	Health, Life
Pro	Health, Life, Accident

Claims cannot be edited after submission.

🏥6️⃣ Claim Rules

Health Claim

- Must be within remaining health coverage
- After approval:
    - Remaining health coverage is reduced
    - Updated balance shown in employee record

Life Claim

- Payout = Salary × Life Multiplier
- Allowed only once
- Employee marked deceased and deactivated

Accident Claim (Pro only)

- Total disability → Full coverage
- Partial disability → Coverage × Disability %

Customer can view claim status (Pending / Approved / Rejected) and remarks.

🔐 Final Authentication & Role Flow

🎯1️⃣ Only One Admin

- One predefined Admin (seeded).
- Admin account is fixed.

- No other Admin can be created.

Correct. Keeps control centralized.

👤2️⃣Who Registers Whom?

🛡️ Admin Registers:

- Agent
- Claims Manager

They do NOT self-register.

🏢 Customer Registers:

- Company registers via website
- Default role = Customer

That is correct and realistic.

🌐3️⃣Single Login Page (Common for All Roles)

You should have:

👉 One Login Page
👉 One Authentication Endpoint

No separate login pages.

Example:

/login

User enters:

- Email
- Password

System authenticates.

🎟️4️⃣ JWT Token Structure

After successful login:

JWT token should include:

```
{
 "userId": "123",
 "email": "abc@company.com",
 "role": "Agent"
}
```

Role can be:

- Admin
- Agent
- ClaimsManager
- Customer

🔀5️⃣ Role-Based Routing After Login

After login:

Frontend reads role from JWT.

Then routes accordingly:

Role	Redirect To

Admin	/admin-dashboard
Agent	/agent-dashboard
Claims Manager	/claims-dashboard
Customer	/customer-dashboard

Very clean.

🛡️6️⃣ Backend Role Authorization (Important)

Each API endpoint must be protected using role-based authorization.

Example in .NET:

[Authorize(Roles = "Admin")]

[Authorize(Roles = "Agent")]

[Authorize(Roles = "ClaimsManager")]

[Authorize(Roles = "Customer")]

This ensures:

- Agent cannot access Admin APIs
- Customer cannot access Claims APIs
- Claims Manager cannot modify policies

Security enforced at backend.

🎯 Final Role Authentication Structure

One Website

- One Login Page
- One JWT System
- Role-based authorization
- Role-based routing
- Separate dashboards

This is the correct enterprise approach.

🏢 Group Employee Insurance Management System

🔄 Complete System Flow

🟢 PHASE 1 — User Setup

1️⃣ Admin Setup

- Predefined Admin exists.
- Admin creates:
  - Agent accounts
  - Claims Manager accounts
- Admin seeds 9 policies (Editable).

🟢 PHASE 2 — Company Registration

2️⃣ Customer Registers

- Company registers as Customer.
- No agent assigned yet.

Agent will be assigned at first policy interaction.

🟢 PHASE 3 — Employee Setup

3️⃣ Company Adds Employees

Customer adds employees with:

- Name
- Email
- Salary
- Role
- Nominee details

Employees can be added before or after policy purchase.

🟢 PHASE 4 — Policy Selection

4️⃣ Customer Views Policies

System shows 9 predefined plans:

Tiers:

- Essential

- Enhanced
- Enterprise

Levels:

- Base → Health only
- Plus → Health + Life
- Pro → Health + Life + Accident

Customer chooses:

Option A — Request Recommendation (Optional)

OR

Option B — Direct Buy

🟢 PHASE 5 — Agent Assignment

If customer performs:

- Recommendation request
  OR
- Direct Buy

System checks:

If AssignedAgentId == null

  Assign Agent

One company → One agent permanently.

🟢 PHASE 6 — Recommendation Flow (If Used)

1. Customer submits recommendation request.
2. Agent sees request in dashboard.
3. Agent selects 3 suitable plans.
4. Agent sends recommendation.
5. Customer views recommended plans.

🟢 PHASE 7 — Quote Request

Customer:

- Selects policy
- Enters employee count
- Clicks “Request Quote”

A Quote Request is created.

Visible to Agent.

🟢 PHASE 8 — Quote Generation

Agent:

- Views selected plan
- Views employee count
- Clicks Generate Quote

System calculates:

$$\text{Premium} = \text{BasePremiumRate} \times \text{EmployeeCount}$$

Agent sends quote to customer.

🟢 PHASE 9 — Policy Activation

Customer:

- Views quote
- Confirms payment

System:

- Activates policy
- Generates Start Date
- Generates Expiry Date (1 year)
- Generates Invoice
- Applies policy to all active employees
- Calculates 8% commission
- Inserts record in Commission table

Only one active policy allowed at a time.

🟢 PHASE 10 — Claim Raising

If employee needs benefit:

Customer:

1. Selects employee
2. Selects claim type (based on policy level)
3. Enters requested amount
4. Uploads documents
5. Submits claim

Status = Pending

Customer cannot edit claim after submission.

🟢 PHASE 11 — Claim Processing

Claims Manager:

- Views claim list
- Filters pending claims
- Clicks claim
- Reviews:
    - Coverage details
    - Remaining coverage
    - Documents
    - Requested amount

Claims Manager:

- Approves OR Rejects
- Adds remarks

🟢 PHASE 12 — Payout Logic

🏥 Health Claim

If RequestedAmount ≤ RemainingHealthCoverage:

- Approve
- Deduct from RemainingHealthCoverage
- Update employee record

💀 Life Claim

- Payout = Salary × LifeMultiplier
- Mark employee IsDeceased = true
- Deactivate employee

🚑 Accident Claim (Pro only)

- Total disability → Full coverage
- Partial disability → Coverage × Disability %

🟢 PHASE 13 — Status Update

After approval/rejection:

- Claim status updated
- Claims Manager stats updated
- Customer notified
- Claim visible in history